

Name: _____ Student ID: _____

HOMEWORK 5

Discrete-Time Fourier Transform

2110203 COMPUTER ENGINEERING MATHEMATICS II

Generative AI Usage Declaration

- ☐ No AI Used: I did not use any AI tools to complete this homework.
- ☐ AI Assisted: I used AI tools as detailed below:

Where	AI model	How

Student Acknowledgement: I declare that I have reviewed, understood, and tested all AI-generated output. I take full responsibility for the accuracy, logic, and functionality of the final submitted work.

Signature: _____ Date: _____

Name: _____

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Write solutions and give explanations (1 point)

SEND ONLY PROBLEM 1. and 2.

1. (0.6 points, 0.2 points each) Use the properties of the discrete-time transform to determine $X(e^{j\omega})$ for the following sequences

1.1. $x[n] = (\frac{1}{3})^{|n|}$

1.2. $x[n] = a^n \cos(\Omega_0 n) \cdot u[n], |a| < 1$

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1.3. $x[n] = (n + 1)a^n \cdot u[n], |a| < 1$

2. (0.4 points, 0.2 points each) Find the discrete-time sequence $x[n]$ with transforms in range $0 \leq \omega \leq 2\pi$ as follows

2.1.

$$X(e^{j\omega}) = -j\pi\delta\left(\omega - \frac{\pi}{3}\right) + \pi\delta\left(\omega - \frac{2\pi}{3}\right) + \pi\delta\left(\omega - \frac{4\pi}{3}\right) + j\pi\delta\left(\omega - \frac{5\pi}{3}\right)$$

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2.2.

$$X(e^{j\omega}) = \frac{1 - \frac{5}{6}e^{-j\omega}}{1 + \frac{1}{12}e^{-j\omega} - \frac{1}{12}e^{-j2\omega}}$$

OPTIONAL, PRACTICE ONLY.

3. Using the DTFT, find the impulse response ($h[n]$) of the system governed by the difference equation
- 3.1. $y(n) = x[n] - 4x[n-1] + \frac{11}{12}y[n-1] + \frac{1}{12}y[n-2]$
- 3.2. $y(n) = x[n] - \frac{11}{15}y[n-1] + \frac{2}{15}y[n-2]$
- and find output with $x[n] = (\frac{1}{2})^n u[n]$.